ISEC 2022

INTERNATIONAL STATISTICAL ECOLOGY CONFERENCE

CAPE TOWN, SOUTH AFRICA — 27 JUNE - 1 JULY 2022

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BioSS

Validating hidden Markov models

to inform seabird conservation

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allaboutbirds.org

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Motivation

- Animal movement is influenced by underlying behaviour

-Inferred behaviours of animals from GPS movement data are rarely validated

Source: movebank

Motivation and Goal

- ## Motivation

  - Seabirds are less protected at sea

  - The need to identify foraging areas of seabirds

  - HMM inferred behaviour of seabirds is rarely validated

- ## Goal

  - Validate the inferred behaviours of seabirds obtained via hidden Markov model

Study species- Terns

- Visually tracked in the UK by Joint Nature Conservation Committee between 2009 - 2011 during

  - incubation (May)
  - chick-rearing (June-July)

- Movement data: GPS coordinates of the boat

- Behavioural data: taken by the observer on the boat

- Data are recorded every second

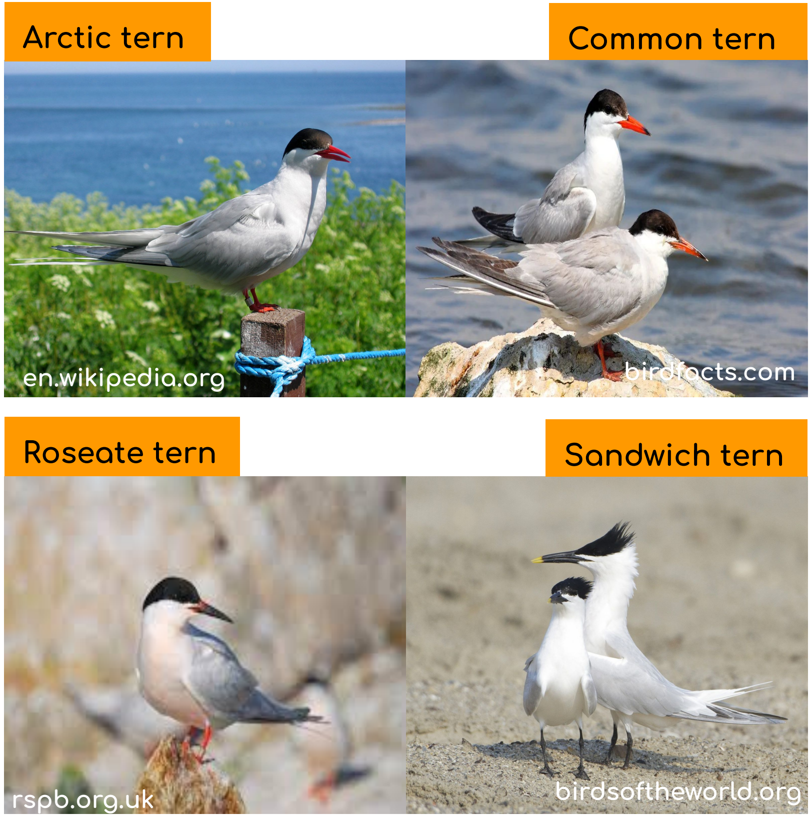


Figure 1: *Species of terns in the UK*

Study site and Data

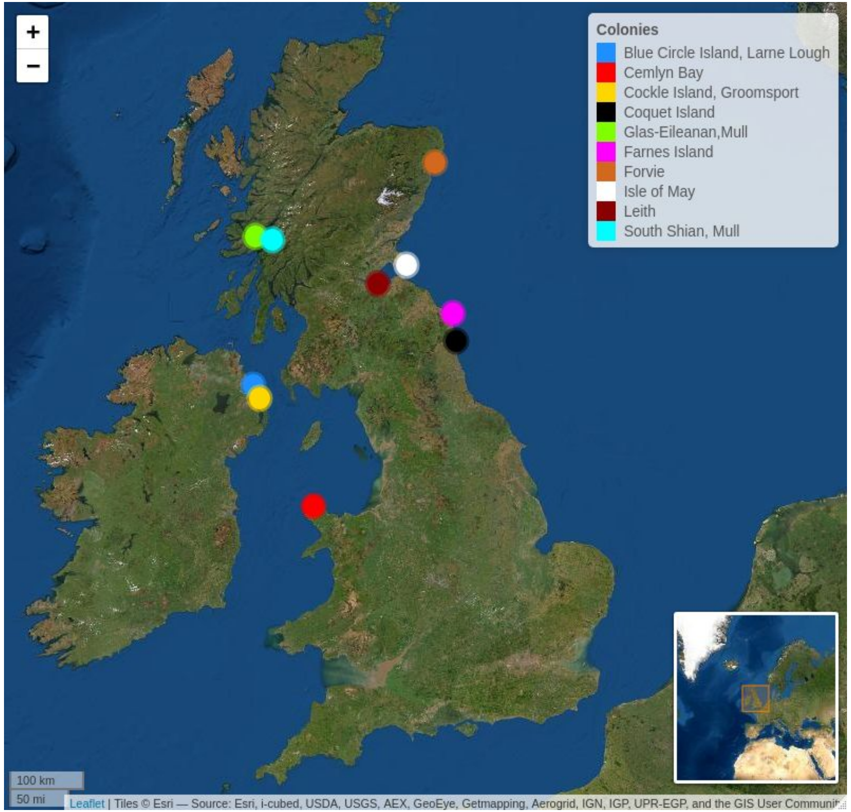


Figure 2: *Study site consisting of 9 tern breeding colonies in the UK*

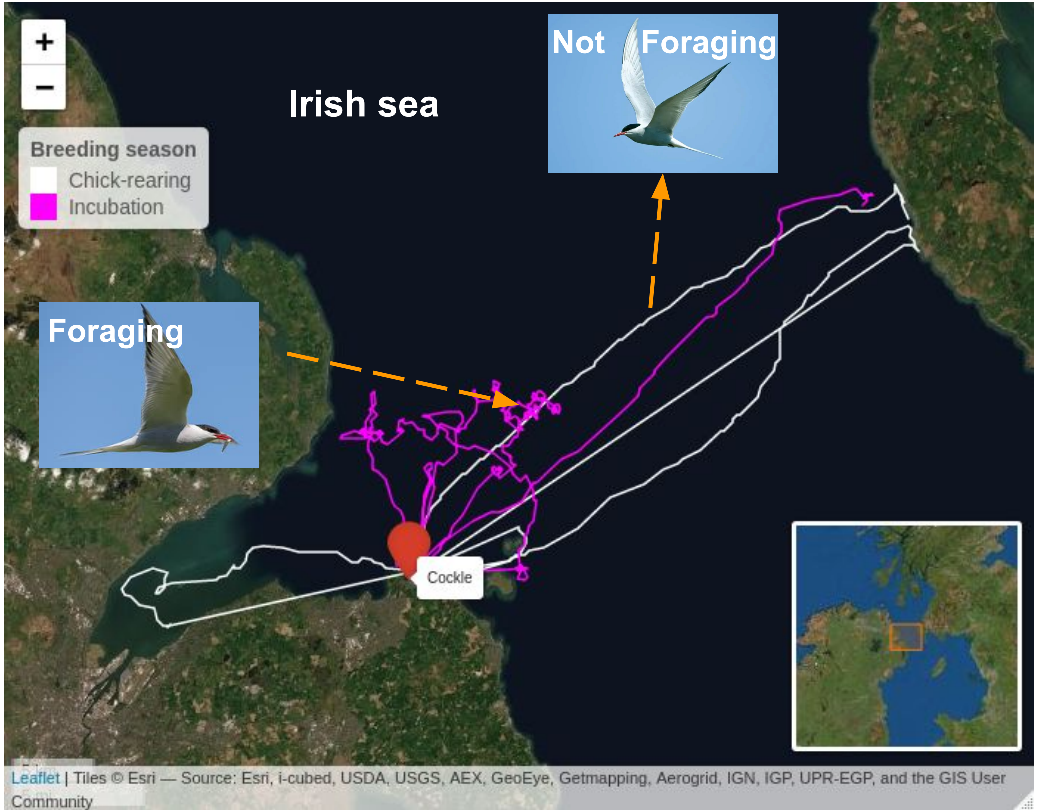


Figure 3: *Visual tracking (boat movement) data of sandwich terns at Cockle Island colony (54°67'N,5°61'W), Northern Ireland, 2010*

Boat movement data as a good proxy

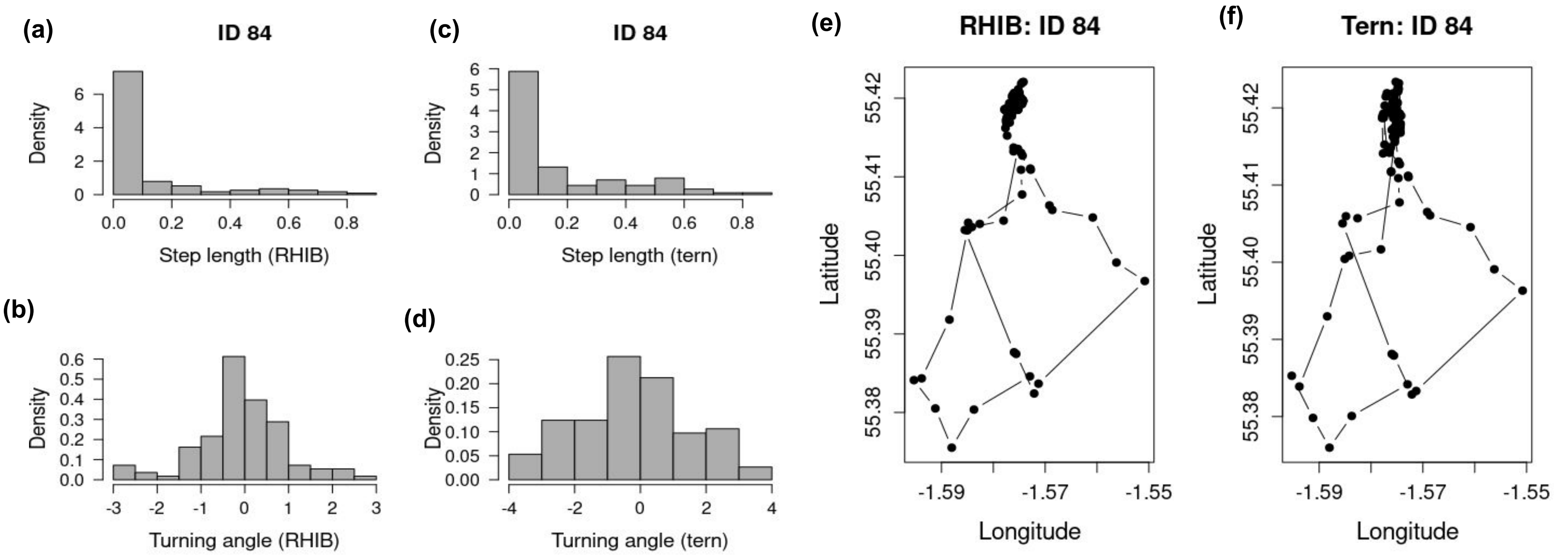


Figure 4: *Histogram of step length and turning angle distribution for visual tracking* ***(a,b)*** *and approximate GPS-tracking* ***(c,d)*** *of roseate tern in Coquet Island colony, 2009. Plot of* ***(e)*** *visual tracking data and* ***(f)*** *approximate GPS-tracking data of a Roseate tern in Coquet Island, 2009.*

Hidden Markov model

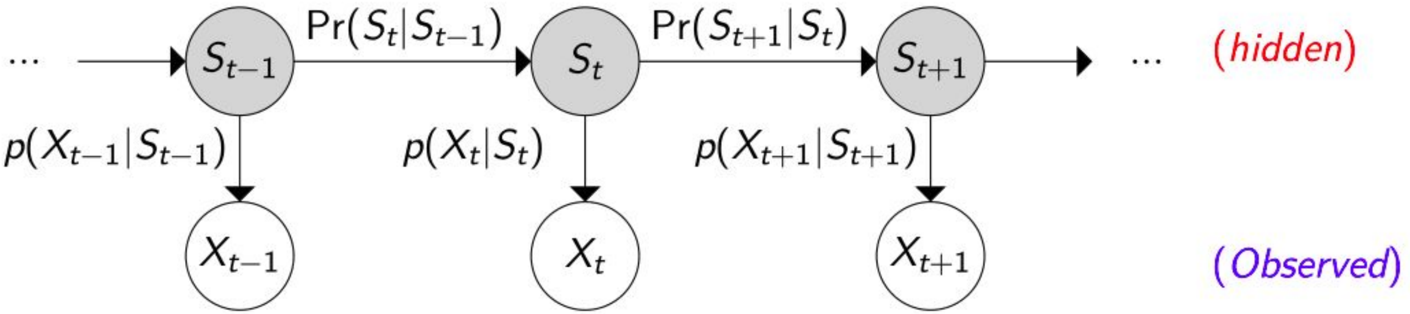


Figure 5: *Representation of a HMM*

- Tern behavioural states: foraging and not-foraging
- Step length and turning angle obtained from proxy (boat) movement data of terns

Validation metrics for HMM

- F1-score

More interested in the positive class: foraging state

$$\text{F1-score} = 2\left(\frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}\right)$$

where

$$\text{Precision} = \frac{\text{No. of TP}}{\text{No. of TP} + \text{No. of FP}},$$

and

$$\text{Recall} = \frac{\text{No. of TP}}{\text{No. of TP} + \text{No. of FN}}.$$

- Logarithmic loss (Log-loss)

Accounts for certainty of how close the decoded state is to the observed state

$$\text{Log-loss}(y, p) = -\left[y \log(p) + (1 - y)\log(1 - p)\right],$$

where

$$y = \begin{cases} 1, & \text{if positive class (observed foraging behaviour)} \\ 0, & \text{if negative class (observed not-foraging behaviour)} \end{cases}$$

and

$$p = \Pr(\text{decoding foraging behavioural state})$$

TP = True Positive, FP = False Positive, FN = False Negative

Result

Table: Summary of HMMs results during chick-rearing period across the 9 breeding colonies. A = Arctic, C = Common, R = Roseate and S = Sandwich tern. Γ = transition probability matrix, covariate = distance to colony.

Chick-rearing		Validation (least log-loss value)		AIC (least value)	
Colony	Species	Preferred HMM	F1-score (%)	Preferred HMM	F1-score (%)
Cemlyn	A	covariate on Γ	76.28	no pool on step length	73.72
Coquet	R	✓	76.38	no covariate	75.08
South Shian	C	✓	79.31	no pool on Γ and step length	77.32
Coquet	C	complete pool*	83.55	✓	83.54
Glas Eileanan	C	✓	78.48	✓	78.56
Leith	C	no pool on step length	72.06	✓	71.76
Blue Circle	S	✓	79.58	✓	79.54
Coquet	A	no pool on Γ	64.19	✓	56.00
Isle of May	A	✓	77.88	no pool on step length	76.24
Cockle	S	no pool on Γ and step length	95.13	✓	95.22
Coquet	S	✓	87.84	no pool on Γ and step length	87.84
Forvie	S	✓	71.77	✓	71.77

* Baseline HMM

Result

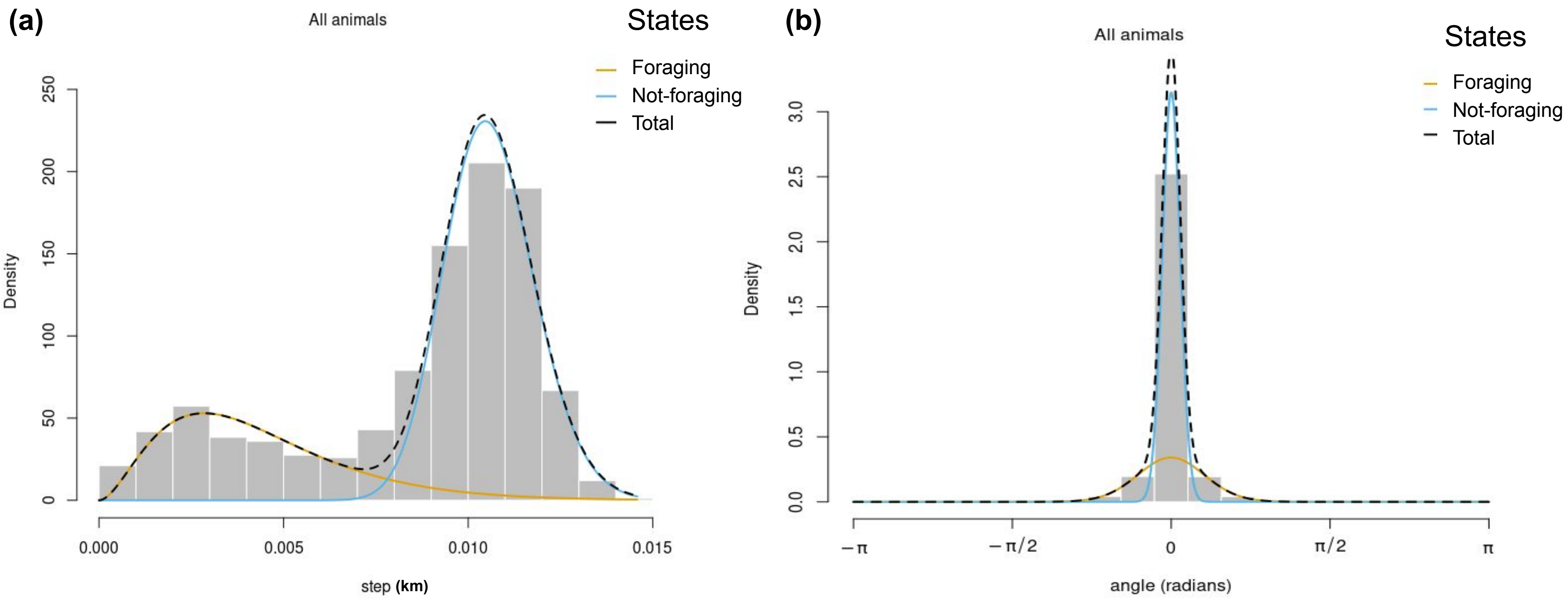
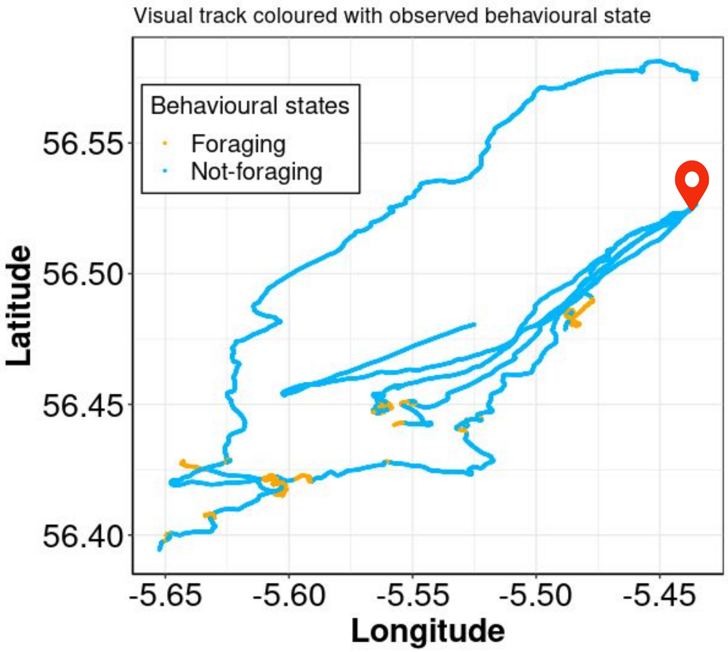


Figure 6: *Histograms of the* ***(a)*** *proxy step length and* ***(b)*** *turning angle distributions of 6 visually-tracked common tern in* ***South Shian colony*** *during chick-rearing period, 2011. Lines represent the HMM fitted state-dependent distributions, and are coloured according to decoded behavioural state:* ***foraging*** *and* ***not-foraging***

Result

(a)



(b)

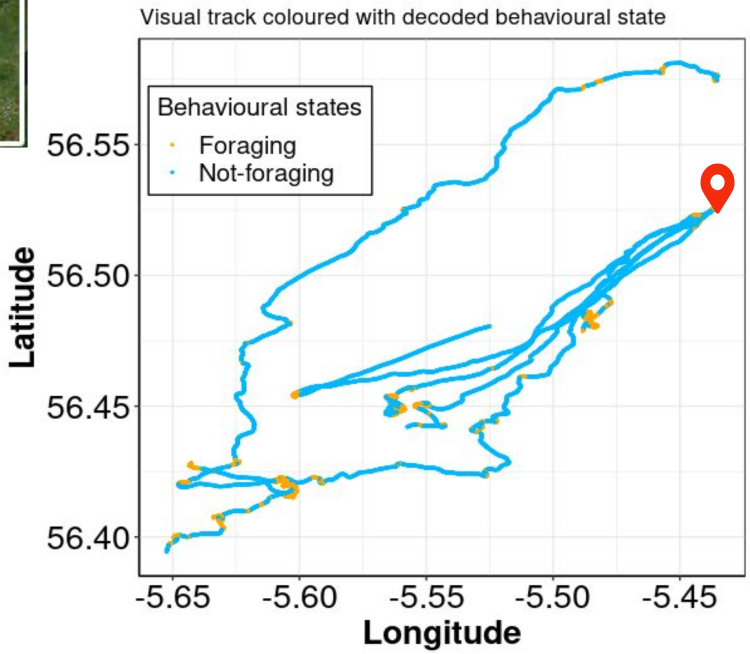


Figure 7: *proxy (boat) movement data of 6 visually tracked common tern coloured with* ***(a)*** *observed and* ***(b)*** *decoded behavioural states:* ***foraging*** *and* ***not-foraging****, in* ***South Shian colony (***📍***)*** *Mull, Scotland, during chick-rearing period, 2011.*

Findings

- Movement data obtained via visual tracking of terns is a suitable proxy to that obtained from GPS devices attached to terns

- Ranking of fitted HMMs based on AIC differs from validation

- Performance of HMMs (based on F1-score) in decoding terns behavioral state ranges between 56 - 95%

Thank you for your attention

Questions?

Image: birdfacts.com